

A Wearable-Driven Physiological Digital Twin for Heart Failure Monitoring: Feasibility and Limits

Kaveri Sharma
Dept. of Computer Science Engg.
PES University
Bengaluru, India
kaveri05sharma@gmail.com

Mrunmayi Mohite
Dept. of Computer Science Engg.
PES University
Bengaluru, India
mrunmayimohite9@gmail.com

Meghan Singhal
Dept. of Computer Science Engg.
PES University
Bengaluru, India
meghansinghal@gmail.com

Adithya Balasubramanyam
CAVE Labs, C-IoT, Dept. of CSE
PES University
Bengaluru, India
adithyakoundinya@gmail.com

Abstract—Consumer wearables measure heart-failure (HF) patients at home but offer no physiological interpretation. We study whether a mechanistic physiology engine can add that layer. Our twin feeds a 21-day window of six wearable trends and routine clinical values to a Random Forest that estimates a deterioration scenario and its severity, simulates the corresponding physiology in the Kitware Pulse engine, and scores the simulated hemodynamics with an interpretable risk score; engine state is carried from day to day. The classifier recovers the labels of its own synthetic generator well (90.7% held-out accuracy), but on a pre-registered cohort of ten scripted 21-day patient stories, day-by-day scenario accuracy was only 17.6%. In an ablation on the same runs, the classifier alone flagged every deteriorating series but alerted in all stable ones (60 false alerts per 100 patient-days); adding the Pulse risk score with a component guard reduced this to 8.7 (95% CI 3.4–16.1) while detecting 70% of deteriorating series, missing fluid overload and deconditioning. A 2 kg-in-3-days weight rule detected 7%. The pipeline ran end to end on real home data from 13 of 16 HF patients, but only through an age-proxy body with default ejection fraction, and the twin’s own scores did not predict real hospital outcomes beyond chance. We conclude that the physiology layer works as a false-alarm filter, not yet as a validated detector, and identify the modeling changes needed next.

Index Terms—Digital twin, heart failure, wearable sensors, physiology simulation, Pulse engine, random forest, remote patient monitoring, early warning, reproducibility.

I. INTRODUCTION

Heart failure affects tens of millions of people worldwide [1], [2] and is defined by progression: patients decline over days to weeks rather than instantaneously. Current guidelines emphasize early recognition of congestion and decompensation [3], yet three structural gaps remain in practice. First, monitoring is episodic: patients are typically seen every few weeks, and a patient can move from stable to crisis between visits. Second, consumer wearables report numbers, not physiology: a rising resting heart rate is measured, but whether it reflects benign variation or an early compensatory response to falling cardiac output is left to the patient. Third, threshold alarms (e.g., $\text{SpO}_2 < 92\%$) fire only after a value is already

abnormal; they describe where a patient is, not where they are heading.

Remote monitoring has shown benefit when it captures physiology rather than symptoms alone, from structured tele-monitoring [4] to implanted pulmonary-artery pressure sensors [5], and multi-sensor wearables combined with machine learning have predicted HF hospitalization days in advance [6], [7]. These data-driven predictors, however, offer little mechanistic explanation of *why* an alert fired. Cardiac digital twins promise exactly that mechanistic layer [8]–[10], but most are built from imaging or invasive measurements fitted to one patient at a time [11], [12], and few are evaluated over weeks of everyday data.

This paper reports a feasibility study of a twin that sits between these two traditions: it is driven by home-measurable wearable trends, interprets them through a mechanistic whole-body physiology engine (Kitware Pulse [14]), and produces interpretable risk, staging and forward projections. We use “twin” in a limited sense: the engine body is parameterized by population statistics and scaled by a patient’s ejection fraction and estimated severity, not fitted to the individual. Our contributions are:

- 1) A complete, open pipeline (wearable trends $\rightarrow$ ML scenario and severity estimation $\rightarrow$ Pulse simulation $\rightarrow$ interpretable risk scoring, staging and projection), deployed as a stateful service with day-to-day continuity of engine state (Sections III–V).
- 2) Engine-level findings for using Pulse as an HF twin: mapping ejection fraction onto the engine, a root-caused instability boundary when disease modifiers and exertion combine, and an exact state save/resume method (Section VI).
- 3) A layered evaluation that includes an ablation isolating what the physiology layer contributes relative to the classifier alone and to a clinical weight rule, on a pre-registered cohort with held-out seeds, plus real-patient replay and real-outcome tests (Section VII).

TABLE I
POSITIONING AGAINST RELATED WORK

Work	Input data	Model	Personalization	Gap addressed here
Corral-Acero 2020 [8], Coorey 2022 [9] Gu 2025 [11]	Multimodal (vision/review) Clinical, echo, invasive	Mechanistic + statistical Lumped cardiovascular model + unsupervised ML	Conceptual Fitted per patient (343 HF)	No deployed system or longitudinal test Requires hospital measurements; no home monitoring
Geddes 2025 [12]	Imaging, invasive validation	3D CFD	Fitted per patient	Right HF only; not wearable-driven
Balasubramanyam 2025 [13] Stehlik 2020 (LINK-HF) [6]	ECG datasets, echo Wearable patch	Compliance model, Fourier ECG, 3D mesh ML (similarity-based)	Signal-level Patient baseline	Visualization and ECG fidelity, not deterioration tracking No mechanistic explanation of alerts
This work	Wearable trends + routine clinical	RF + Pulse engine + interpretable score	Scenario/severity-driven, EF-scaled	Mechanistic, wearable-driven, evaluated over 21 days

II. RELATED WORK

Cardiac digital twins. Corral-Acero *et al.* [8] framed the digital twin as the route to precision cardiology, Coorey *et al.* [9] reviewed the emerging health-twin field for cardiovascular disease, and Niederer *et al.* [10] argued that twins must move from artisanal, single-patient builds to scalable, industrial pipelines. Recent HF-specific twins fit mechanistic cardiovascular models to measured data: Gu *et al.* [11] fitted twins to 343 HF patients and used the fitted parameters for interpretable phenotyping and prognosis, and Geddes *et al.* [12] built CFD-based twins that non-invasively estimate markers of right HF. Balasubramanyam *et al.* [13] proposed a cardiovascular twin combining time-varying compliance, Fourier-based ECG reconstruction and 3D anatomical modeling.

Wearable HF prediction. The LINK-HF study [6] combined a wearable patch with machine learning to predict HF hospitalization with a median lead time of several days, and Gautam *et al.* [7] surveyed AI, wearables and remote monitoring in HF. Invasive hemodynamic monitoring reduced HF hospitalization in the CHAMPION trial [5].

Risk scores. The MAGGIC score [15] is a widely used, externally validated HF survival score, and the LACE index [16] is a general readmission score; both serve as external benchmarks here.

Positioning. Table I contrasts these approaches. Mechanistic twins personalize deeply but require hospital measurements; wearable predictors scale but are opaque. Our twin trades per-patient parameter fitting for deployability from home data, and adds a mechanistic layer that wearable-only predictors lack. We claim novelty for the integrated pipeline and its evaluation, not for any single component.

III. SYSTEM OVERVIEW

Fig. 1 shows the pipeline. A patient record holds demographics and a clinical report (ejection fraction, NT-proBNP); daily wearable readings accumulate until a 21-day window is filled, which triggers one background assessment: (1) the scenario classifier and severity regressor (“Model 1”); (2) construction of a Pulse patient and scenario file; (3) simulation inside the Pulse container with crash and timeout detection;

TABLE II
TECHNOLOGY STACK

Layer	Component
Physiology engine	Kitware Pulse 4.3.1 (kitware/pulse:4.3.1 image, amd64) [14]
Machine learning	scikit-learn 1.9.0 Random Forest [17], [18]; XGBoost 3.2.0 (secondary) [19]
Back end	Python 3.11, FastAPI, SQLAlchemy, Pydantic; SQLite (Postgres-portable)
Front end	React 19 + Vite; hand-built SVG charts
Deployment	Docker Compose (Postgres, back end, front end)
Testing	pytest (281 automated tests), Playwright browser checks, GitHub Actions CI

(4) hemodynamic feature extraction; (5) risk scoring, NYHA staging and trend-based deterioration rate; and (6) three further Pulse runs at projected severities for 7, 14 and 30 days. All results are persisted, so every read endpoint is a database read and never blocks on the engine. Table II lists the technology stack.

IV. DATA

A. Reference data and synthetic training population

No real patient data were used for training. Instead, every distribution parameter used by the synthetic generator is traced to an aggregate statistic from a real source (Table III): MIMIC-IV HF admissions [20] for age, sex and laboratory values; the Chicco–Jurman HF clinical records [22] as the only source with ejection fraction; and NHANES for height, weight and BMI by sex. A generator produced $n = 2,000$ synthetic patients across five scenarios. Each patient receives a scenario and a severity $\sigma \sim U(0, 1)$ (capped at 0.15 for stable patients). Ejection fraction is drawn from the scenario’s EF profile and lowered by 12σ points; NT-proBNP is driven by a mix of EF deficit and severity scaled to age-specific diagnostic cut-offs, which makes BNP fall with EF and rise with severity. Each wearable signal v on day t then follows

$$v(t) = b_v + \Delta_{v,c} \sigma g_m(t/20) + \epsilon_{v,t}, \quad (1)$$

where b_v is a patient baseline (drawn from population statistics, or the patient’s own weight), $\Delta_{v,c}$ is the maximum

TABLE V
RISK-SCORE COMPONENTS AND CLINICAL ANCHORS

Component	Weight	Anchor (citation)
HR rise	0.15	NEWS2 heart-rate bands [26]
MAP drop	0.20	Healthy baseline to MAP 65 mmHg [27]
CO drop	0.20	30% decline = full scale [28], [29]
Compensation failure	0.15	Frank-Starling failure (binary)
Instability flag	0.30	MAP < 65 mmHg [27]
Baseline deficit	max()	Starting MAP below 92.5 mmHg anchor

TABLE VI
WHEN EACH SCORING ELEMENT WAS FIXED

Element	Status	Independent evaluation
Risk weights and anchors	Pre-specified (literature)	All later data
Bucket cut-points 0.35/0.65	Pre-specified (engineering)	All later data; not clinically validated
Baseline-deficit term	Post-hoc on 117-run batch	Scenario test, PerHeart run 3, MIMIC-IV, Zigong
NYHA feature removed	Post-hoc on live batch	45 new live patients; PerHeart
C3 component guard	Post-hoc on dev seeds	Held-out seeds 45–47
WATCH tier (decide_alert)	Post-hoc	Replay on both seed sets only
Scenario expectations	Pre-registered	—

C. Hemodynamic features and risk score

From each run we extract HR rise, MAP drop, cardiac-output (CO) drop, a compensation flag (stroke volume held $\geq 95\%$) and an instability flag ($\text{MAP}_{\text{end}} < 65 \text{ mmHg}$). The primary risk score (Table V) normalizes each to $[0, 1]$ against a clinically anchored full scale and forms a weighted sum. During validation the score was structurally blind to fluid overload, whose danger appears as an already-congested baseline rather than an acute change, so a sixth input was added:

$$r = \max \left(\sum_i w_i s_i, s_{\text{baseline}}(\text{MAP}_{\text{start}}) \right), \quad (2)$$

where s_i are the five acute sub-scores and s_{baseline} scores how far the simulated starting MAP lies below a healthy 92.5 mmHg anchor. Buckets are LOW ($r < 0.35$), MODERATE ($r < 0.65$) and HIGH; these cut-points are an engineering choice, not a clinical citation. An XGBoost regressor [19] on the same features is kept as a secondary comparison only (5-fold CV MAE 0.089 ± 0.020 , R^2 0.828 ± 0.091 ; $n = 117$ is too small to make it primary).

D. Pre-specified and post-hoc design decisions

Several scoring elements were changed after seeing results. Table VI records when each was fixed relative to the data it was evaluated on, so that results on data used for tuning are not mistaken for out-of-sample evidence.

E. Staging, deterioration rate and projection

NYHA staging is rule-based: AHA/ACC structural criteria ($\text{LVEF} \leq 40\%$ or age-adjusted NT-proBNP above cut-off) [3]

gate whether symptoms have a structural basis, and the simulated exertion response places the patient in classes I–IV. A deterioration rate is computed from per-vital slopes normalized to population standard deviations per day; projection re-runs the full Pulse pipeline at linearly extrapolated severities for 7, 14 and 30 days.

F. Service, alerts and continuous state

The FastAPI service exposes eight endpoints over five tables (patients, clinical reports, wearable readings, simulation runs, risk assessments). Missing EF or NT-proBNP defaults to a documented healthy value (EF 62%, NT-proBNP 100 pg/mL), always flagged and never silently blended with measured values. A continuous-state mode serializes the Pulse engine state after each day and resumes from it the next day, instead of rebuilding the patient from scratch (Section VI).

VI. ENGINEERING CHALLENGES AND SOLUTIONS

Using Pulse as an HF twin surfaced problems that were not visible from documentation or offline evaluation. Table VII lists those specific to the engine and the modeling; general software and infrastructure issues are in Appendix Table XIX.

A. Pulse instability at high severity

In a 150-patient batch (30 per scenario), 117 runs completed (Fig. 2). All 33 failures occurred in the two scenarios that add an `Exercise` action; 30 were engine crashes and 3 were timeouts, and failed patients had high severity (mean 0.75 and 0.72), so missingness is *not at random* with respect to severity. Engine logs for three independent crashes showed an identical chain: fatigue, hypoxia and tachycardia, then renal hypoperfusion and cardiovascular collapse, ending in a fatal negative heart-chamber volume and an irreversible-state event at 154.1–154.3 s. Table VIII lists single-variable interventions on a reproducible case. Gaps and ramps only delayed the crash; either stressor alone completed. The failure is therefore a sustained combined-load boundary whose safe exercise intensity varies by patient (about 0.22 for one patient and between 0.125 and 0.25 for another), so no single safe constant exists.

B. Two failure mechanisms with opposite character

Failures were not a single phenomenon. Mechanism 1 (engine crash) depends on the patient’s own severity and is deterministic, so retrying an identical scenario does not help. Mechanism 2 (timeouts and connection-pool exhaustion) depended on how many other patients were running: at 9 workers, 5/11 patients hit connection timeouts and 6/11 hit the 180 s engine timeout; at 4 workers 8/11 still timed out; at 2 workers there were no failures. A bare completion rate conflates the two; we report the mechanism with each rate.

C. Continuous state: carrying a patient forward

Rebuilding a patient every day discards the physiological state the twin is meant to track. Pulse can serialize its full state (about 2.33 MB, saved in 0.31–0.44 s and loaded in 0.43 s) and

TABLE VII
ENGINE AND MODELING PROBLEMS, ROOT CAUSES AND SOLUTIONS

Problem	Root cause (how found)	Solution	Evidence
Pulse rejected scenario files	Actions/conditions need PatientAction/AnyCondition wrappers; found from protobuf parse errors	Builders emit the wrapped schema	All 5 scenarios run
Runs exited 1 with time mismatch	CardiovascularMechanicsModification without Incremental:true silently re-stabilizes, consuming extra time	Set Incremental:true	Clean exit, exact end time
Collapse at low EF (negative chamber volume)	Binary EF condition stacked with a second EF-derived cut	Continuous multiplier carries only the additional acute severity	Collapse no longer occurs
Real patients outside Pulse range	Engine hard-rejects age > 65, BMI > 30; real HF cohorts are older	Clamp demographics only at the engine boundary; flag per patient	16/16 PerHeart patients flagged
Deconditioning looked like cardiac stress	Both used Exercise; HR 150–164 in both	Removed exertion from deconditioning	Distinct HR/MAP profiles (Table XII)
Fluid overload always scored LOW	Risk only measured change within a run; fluid overload is a shifted baseline	Added baseline-deficit term with max()	30/30 LOW → 29/30 MODERATE
Live severity error 0.271 vs. 0.048 offline	NYHA ordinal used in training but unknown at inference, defaulted to class I	Removed NYHA from features; retrained	Live MAE 0.271 → 0.026
Engine crashes at high severity	Disease modifiers + sustained exertion exceed engine’s sustainable load (Table VIII)	Characterized; failures recorded, never dropped	Deterministic at 154.1–154.3 s
Resumed state drifted	Disease modifiers are not carried by serialized state	Re-issue modifiers on every resume	Drift 6.5% → 0.000%
Engine segfault in continuous mode	Pulse Python SDK initialize_engine() crash (gdb trace); earlier “thread conflict” diagnosis disproved	Switched to the CLI scenario driver	Day 1→3 runs clean
Resumed runs flagged as crashes	Driver’s end-time check ignores elapsed state time; schema field renamed (Mode, not Type)	Resume-aware crash filter; corrected schema	Exact match vs. continuous
SpO ₂ output always 0.0	Persists with Pulse’s bundled patient and a request byte-identical to a known-good one; likely arm64 emulation	Never used as a feature	Excluded
ECG looked computed	Pulse ECG is a stored template scaled by HR	Labeled display-only in UI/API	Caveat in every response

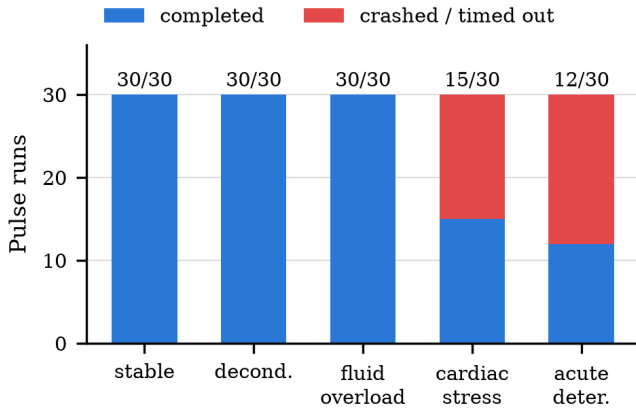


Fig. 2. Pulse batch completion by scenario (150 runs). All failures occur in the two scenarios that combine disease modifiers with an Exercise action.

resume it, and with an Exercise action resumption was bit-for-bit identical to an uninterrupted run. With the disease modifier used by our pipeline it was not (Table IX): the modifier is not carried by the serialized state. Re-issuing it on every resume restored an exact match, which also requires storing the last-used EF and severity alongside the state. A later end-to-end test crashed inside the Pulse Python SDK; isolating it showed a native crash in the SDK’s engine initialization even with no

TABLE VIII
ISOLATION EXPERIMENTS ON A REPRODUCIBLE CRASH (SEVERITY 0.731)

Intervention	Outcome
Control (exact reproduction)	Crash at 154.32 s
60 s gap before exertion	Crash at 212.02 s
120 s gap before exertion	Crash at 272.92 s
Exertion ramped in 4 steps	Crash at 301.14 s
Exertion alone, no disease modifiers	Completes (660 s)
Disease modifiers alone, no exertion	Completes (660 s)
Exertion intensity halved	Completes (660 s)

TABLE IX
STATE RESUME FIDELITY (PATIENT EF 56.4, SEVERITY 0.46; FINAL VALUES AT 240 s)

Run	HR	MAP	CO	SV
Continuous (reference)	74.13	95.22	6.145	82.89
Resumed, modifier not re-issued	71.51	95.34	5.743	80.32
difference	−3.5%	+0.1%	−6.5%	−3.1%
Resumed, modifier re-issued	74.13	95.22	6.145	82.89
difference	0.000%	0.000%	0.000%	0.000%
Two continuous runs (control)	identical			

machine-learning code loaded, disproving an earlier thread-conflict diagnosis, so the feature was rebuilt on the command-line driver.

TABLE X
EVALUATION LAYERS AND THEIR SCOPE

Layer	Data	Main result	Does not test
Offline model test	300 held-out synthetic patients	90.7% accuracy, severity MAE 0.047	Real patients
Engine scenario check	5 synthetic patients in Pulse	Physiologically distinct scenario responses	Clinical accuracy
Live API validation	25 then 45 synthetic patients via live API	25/25 completed; live MAE 0.026 after fix	Real patients
Pre-registered scenario test	10 scripted stories $\times$ 6 seeds $\times$ 21 days	3/5 stories detected in all seeds; label accuracy 17.6%	Real deterioration events
Ablation (same runs)	30 deteriorating, 18 stable series	Physiology layer cuts false alerts 60 $\rightarrow$ 8.7 per 100 patient-days; sensitivity 100% $\rightarrow$ 70%	Clinician interpretability
Real-patient replay	PerHeart, 16 HF patients	13/16 completed end to end	Outcomes; patients' own physiology (age proxy)
BCG hemodynamics	2 patients, measured SV/CO	Simulated CO 45–46% below measured	Cohort-level accuracy; trends
Real-outcome tests	MIMIC-IV, Zigong	Twins' own scores near chance	Day-by-day early warning

D. Other modeling corrections

Three further corrections followed from running the real system rather than offline checks. (i) Severity (classifier output) and risk score (simulation-derived) share a $[0, 1]$ range but not a scale: for acute deterioration the risk score's minimum (0.491) exceeded the severity mean (0.385). The projection module had applied a risk-score calibration to severity; the two calibrations were separated and the two quantities are now reported side by side, never fused. (ii) The classifier's label occasionally flipped between days for genuinely ambiguous trajectories; a categorical persistence rule (window chosen by sweep) now stabilizes the reported scenario. (iii) An attempt to add rolling-window curvature features ("Option A") kept accuracy flat (90.7%) but worsened severity MAE from 0.0473 to 0.0506 on held-out data, so it was reverted.

VII. EVALUATION

We evaluate in layers of increasing realism. Table X summarizes what each layer tests and what it does not.

A. Offline model accuracy

On the 300-patient held-out test set, the classifier reached 90.7% accuracy (macro-F1 0.91) and the regressor severity MAE 0.047 (RMSE 0.061). Because test patients come from the same generator, these figures measure how well the model inverts the generator, not how it performs on independent trajectories; the pre-registered cohort below (17.6% day-by-day label accuracy) is the more relevant estimate. Cardiac stress (preserved EF) and acute deterioration (reduced EF), the pair hardest to separate from simulated HR/MAP alone, were cleanly separated because EF is an input. Removing all 24 wearable features cost 40 points of accuracy (Table XI), confirming that the wearable trend, not the clinical snapshot, carries most of the signal. The earlier 92.3% figure belonged to a version that included NYHA class, removed for the reason in Table VII.

B. Engine-level scenario validation

Each scenario was simulated for one patient at severity ≈ 0.5 (Table XII). Responses were physiologically distinct. The clearest contrast is between cardiac stress and acute

TABLE XI
MODEL 1 ON 300 HELD-OUT SYNTHETIC PATIENTS

Variant	Accuracy	Severity MAE	RMSE
With NYHA feature (superseded)	92.3%	0.048	0.063
Production (clinical + wearable)	90.7%	0.047	0.061
+ curvature features (reverted)	90.7%	0.051	0.065
Clinical features only	50.3%	0.171	0.219

TABLE XII
SIMULATED RESPONSE BY SCENARIO (SEVERITY ≈ 0.5)

Scenario	HR (bpm)	MAP (mmHg)	Note
stable	71 $\rightarrow$ 70	95 $\rightarrow$ 95	Flat
deconditioning	71 $\rightarrow$ 74	95 $\rightarrow$ 95	Mild drift
fluid_overload	72 $\rightarrow$ 72	77 $\rightarrow$ 77	Congested baseline; CO +13%
cardiac_stress	71 $\rightarrow$ 164	95 $\rightarrow$ 67	Healthy compensation (EF 67%)
acute_deter.	72 $\rightarrow$ 132	78 $\rightarrow$ 52	SV cannot rise (EF 26.6%)

deterioration: both raised heart rate, but the healthy heart (EF 67%) raised cardiac output from 5,910 to 9,948 mL/min with stroke volume maintained, while the failing heart (EF 26.6%) could not increase stroke volume (63 $\rightarrow$ 67 mL) and MAP fell to 52 mmHg. This contrast was not tuned; it emerged from the EF-driven modifiers. The engine's left-heart pressure-volume signal was also physiologically plausible (volume 62.2–143.0 mL, pressure 5.8–136.8 mmHg at EF 45%) and is shown on the dashboard as a PV loop.

C. Live API validation

Twenty-five synthetic patients (five per scenario) were replayed day by day through the live API with no mocking: 25/25 completed and all 25 were classified correctly, but severity MAE was 0.271, far from the offline 0.048. Per-patient inspection showed predictions compressed into a narrow 0.02–0.19 band, traced to the NYHA train/inference skew. After removing the feature and retraining, live re-validation reached severity MAE 0.008 on five patients and, expanded to 50 attempted (45 completed), MAE 0.0264 (95% bootstrap CI 0.0194–0.0352) with 100% scenario accuracy. The five

TABLE XIII
PRE-REGISTERED SCENARIO COHORT

ID	Story	Group
P01	Stable patient, ordinary life	stay quiet
P02	Salty weekend (weight bump, recovers)	edge case
P03	Slow, quiet weight gain	edge case
P04	Fluid overload building (sev. 0.20→0.50)	catch
P05	Gradual deconditioning (sev. 0.20→0.35)	catch
P06	Cardiac stress, exertion episodes	catch
P07	Sudden deterioration, days 10–13	catch
P08	Stressful fortnight, healthy heart	stay quiet
P09	Active, stable patient	stay quiet
P10	Everything goes wrong (sev. 0.20→0.45)	catch

failures were all high-severity exertion scenarios, matching Mechanism 1.

D. Pre-registered 21-day scenario test

Ten patient stories were written and their expected outcomes committed before any simulation (Table XIII): five should be caught, three should stay quiet and two are edge cases. Each story was run for 21 days in continuous-state mode on development seeds 42–44 and, after a rule was selected, on held-out seeds 45–47. A pilot run exposed a design confound (EF matched to story severity) and two harness bugs (day-to-day noise scaled to the current value instead of the population SD, and per-process salting of Python’s `hash()` making seeds non-reproducible); all were fixed and the harness was frozen before the main runs, with every amendment recorded.

Two alert definitions. The first analysis alerted on the classifier’s severity (> 0.15) and caught all five deteriorating patients, but also alerted on all three stable ones (58.7 false alerts per 100 patient-days). A code audit then showed that the dashboard did not use this signal at all: it displayed the risk-score bucket (HIGH). Re-analysed under the dashboard’s definition, the system detected three of five deteriorating patients (P06, P07, P10) with 18.0 false alerts per 100 patient-days. P05 (deconditioning) produced near-zero values on every risk component on all 63 patient-days, and P04 (fluid overload) did not congest in some seeds, so neither had any signal for an alert rule to act on.

Alert rule selection. Three candidate rules were specified before computing any of them (Table XIV). C3 downgrades a sustained HIGH that is driven only by the baseline-deficit term with no corroborating instability, the diagnosed cause of P08’s persistent false alarm. It kept all three detections and cut false alerts by 67.6% on development seeds and 61.4% on held-out seeds, where it was never tuned.

Unified alert decision. The two alert signals were then merged into a single `decide_alert()` function, which is now the only source of alert state in the API and dashboard. It has three levels: ALERT (risk bucket HIGH and not downgraded by C3, or a crashed/failed engine run via the classifier-severity fallback); WATCH (risk bucket MODERATE, or a HIGH streak that C3 downgrades); and NONE. C3 therefore downgrades but never silences: a HIGH risk bucket can never map to NONE. Replaying the saved scenario-test runs through

TABLE XIV
ALERT RULES: DETECTIONS AND FALSE ALERTS

Rule	Dev. (42–44)		Held-out (45–47)	
	Caught	FA/100	Caught	FA/100
Baseline (risk bucket HIGH)	3/5	17.99	3/5	30.16
C1: HIGH for 3 consecutive days	3/5	13.76	3/5	25.40
C2: add sustained-MODERATE watch	3/5	17.99	3/5	30.16
C3: component guard	3/5	5.82	3/5	11.64

TABLE XV
ALERT-SYSTEM ABLATION ON THE SAME RUNS (ALL SIX SEEDS)

Alert system	Sensitivity	Specificity	FA/100 pd
Weight rule	2/30 (0.02–0.21)	18/18 (0.82–1.00)	0.0
Classifier only	30/30 (0.89–1.00)	0/18 (0.00–0.18)	60.1 (44.4–73.8)
+ Pulse risk (HIGH)	21/30 (0.52–0.83)	10/18 (0.34–0.75)	24.1 (10.6–38.6)
+ C3 guard (deployed)	21/30 (0.52–0.83)	10/18 (0.34–0.75)	8.7 (3.4–16.1)
Deployed, WATCH or ALERT	22/30 (0.56–0.86)	9/18 (0.29–0.71)	29.1 (14.6–46.3)

Wilson 95% CIs for proportions; bootstrap 95% CIs (2,000 resamples over series) for false alerts per 100 patient-days in the stable stories. Development seeds alone: deployed system 5.8 (1.6–11.6); held-out: 11.6 (2.6–24.3).

the deployed functions reproduced the detection pattern on both seed sets (P06, P07 and P10 at ALERT in every seed; P01 and P09 quiet in every seed). P08’s false alarm, previously ALERT for 8–14 consecutive days, was reduced to about 3 ALERT days per seed, with the remaining elevated days shown as WATCH. For P04 held-out seed 47, which never reached HIGH, 20 of 21 days now read WATCH, making a previously invisible fluid-overload trajectory visible for the first time.

Ablation: what does the physiology layer add? The premise of the twin is that simulating physiology adds something a wearable classifier alone does not. We tested this on the same saved runs by changing only the alert definition (Table XV, Fig. 3). Series-level sensitivity counts deteriorating patient-seed series with at least one alert on or after the story’s onset day; specificity counts stable series with no alert at all. The classifier alone (severity above the stable range) detected all 30 deteriorating series but alerted in every one of the 18 stable series. Passing the same estimate through Pulse and the risk score removed most of those alerts, and the C3 guard removed most of the rest: false alerts fell from 60.1 to 8.7 per 100 patient-days, while sensitivity fell from 100% to 70%. The physiology layer therefore acts as a filter that rejects alerts unsupported by simulated hemodynamic stress; it does not detect anything the classifier misses. A conventional weight rule (≥ 2 kg within 3 days) never false-alerted but detected only 2/30 series, because the scripted fluid build-up (about 0.4 kg per day) never reaches 2 kg within three days.

Absolute burden and sample size. The relative reductions hide a clinically heavy absolute rate: 8.7 false alerts per

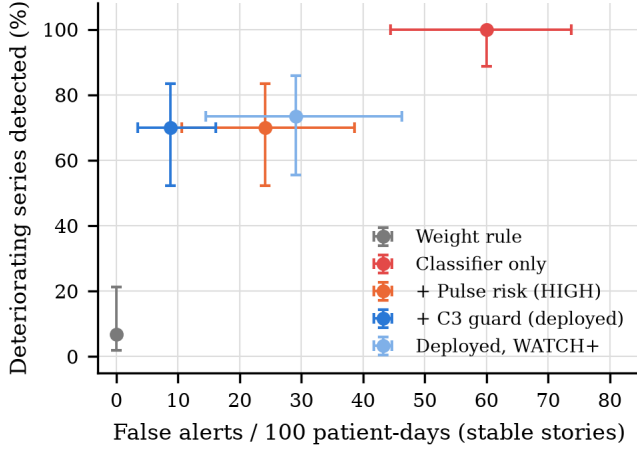


Fig. 3. Sensitivity versus false-alert burden for each alert definition, applied to the same saved runs (95% CIs). The physiology layer moves the classifier from the top right toward the top left.

TABLE XVI
SENSITIVITY OF PREDICTED SEVERITY TO ADDED WEARABLE NOISE

Patient	0×SD	1×SD	1.5×SD	2×SD
P01 (stable)	0.115	0.428	0.544	0.560
P10 (deteriorating)	0.777	0.570	0.622	0.850

100 patient-days means roughly one false alert every 11 or 12 days for each stable patient, and one every 9 days on the held-out seeds alone. With ten stories and six seeds, the confidence intervals are wide (Table XV); these results should be read as a controlled case study, not as estimates of performance in a patient population. Engine failures did not distort them: only 3 of 1,260 patient-days failed, and every metric was identical whether failed days were scored as alerts (the deployed fallback) or as no alert.

Why false alerts occur. Feature importances show the severity regressor relies on resting-HR delta and slope (63.2% combined) and step-count trends (27.2%), the signals that ordinary day-to-day variation moves most, while EF ranks tenth (0.6%). Adding Gaussian noise at one population SD to a stable patient’s window moved predicted severity from 0.11 to 0.43, crossing the stable-range threshold (Table XVI). A follow-up analysis also found that the cohort’s neutral EF band (48–57%) sits below the training “stable” class’s EF range (51.3–70.7%) and near the deconditioning mean, so day-by-day scenario-label accuracy on this cohort was only 17.6% (222/1,260 patient-days), against 90.7% on the in-distribution test set.

E. Real-patient replay (PerHeart)

The PerHeart pilot dataset [23] contains one month of real home measurements (pulse oximeter, scale, blood-pressure cuff) from 27 HF patients aged 67–94. Sixteen had at least 21 real overlapping days of heart rate, SpO₂ and weight; no day was fabricated, and unmeasured fields (sleep, HRV,

TABLE XVII
SIMULATED VS. MEASURED HEMODYNAMICS (RATIO SIM/REAL)

Subject	HR	Stroke volume	Cardiac output
14	0.94	0.58	0.54
102	0.63	0.87	0.55

steps, EF, NT-proBNP) took documented defaults with explicit flags. Every patient required the engine’s age proxy. Before the severity fix, all 16 completed but severities were compressed (0.078–0.148) and no patient reached HIGH, independently reproducing the NYHA skew on real data. After the fix, severities spread (0.093–0.516) and risk buckets were 4 LOW, 4 MODERATE and 5 HIGH (previously 4 LOW, 12 MODERATE, 0 HIGH), while three patients now crashed with the deterministic engine signature. All 16 were staged NYHA I, an artifact of the identical healthy EF/BNP defaults rather than a clinical finding. The one fluid-overload patient scored LOW because the EF default makes Pulse simulate a structurally normal heart, so the baseline-deficit term cannot fire; the API now names this mechanism in its caveat. We searched the literature for a BNP-to-EF proxy and found none: the clinical relationship runs from EF to BNP, not the reverse.

F. Measured hemodynamics (BCG, two patients)

For two patients from a ballistocardiography dataset with echocardiographic volumes [25], BCG features adjusted the twin’s vascular compliance and simulated stroke volume and cardiac output were compared with measured values (Table XVII). Simulated cardiac output was 45–46% below measured in both; 71.3% of patient 14’s stroke-volume shortfall was already present from anthropometrics alone, before EF or BCG data entered the model. A synthetic 21-day deterioration anchored to these patients’ real HR and EF was eventually classified correctly for both, but patient 14 was labeled fluid overload for days 6–10 before converging.

G. Real-outcome tests

Two real cohorts lack the 21-day wearable window and EF needed to run the whole twin, so only components with a non-fabricated input path were tested (Fig. 4, Table XVIII). The baseline-deficit term is a function of starting MAP alone; applied to measured first-24-hour MAP in 17,129 MIMIC-IV HF ICU admissions, it predicted in-hospital mortality with AUC 0.596 (95% CI 0.585–0.608), with mortality rising from 8.9% to 21.2% across score deciles. This shows that low admission MAP carries modest prognostic information in an ICU population; it is not a test of the twin, which was not run on these patients. On the Zigong cohort [24] (6-month death or readmission), the same term and a clinical-only Model 1 variant were indistinguishable from chance. As external context, published LACE-index AUCs [16] lie around 0.56–0.65 in other populations and outcomes (not directly comparable), an 11-of-13-variable MAGGIC score reached 0.604 on this cohort, and a model using laboratory panels

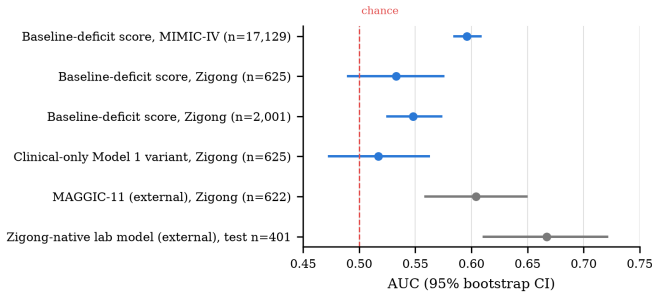


Fig. 4. Real-outcome discrimination. Blue: components of the twin. Grey: external benchmarks on the same cohort.

TABLE XVIII
REAL-OUTCOME TESTS (AUC, 95% BOOTSTRAP CI)

Score / cohort	<i>n</i>	AUC
Baseline deficit, MIMIC-IV, in-hospital death	17,129	0.596 (0.585–0.608)
Baseline deficit, Zigong, complete-case	625	0.533 (0.490–0.575)
Baseline deficit, Zigong, all cleaned	2,001	0.548 (0.525–0.573)
Clinical-only Model 1, Zigong	625	0.517 (0.473–0.562)
MAGGIC-11 (external) [15], Zigong	622	0.604 (0.559–0.649)
Zigong lab model (external), test set	401	0.667 (0.611–0.721)

Zigong outcome: death or readmission within 6 months (41.3% event rate). Bootstrap: 2,000 resamples.

reached 0.667, signal carried by renal and coagulation tests that a wearable-based system cannot obtain.

On the synthetic batch, the risk score agreed moderately with MAGGIC computed on the same patients (Spearman $r = 0.479$; tier agreement 38.5% against $\approx 33\%$ by chance), indicating directional consistency rather than validity.

VIII. DISCUSSION

What the twin does well. The pipeline runs end to end, day after day, on synthetic and real home data; it separates physiologically different presentations through mechanism rather than labels (Table XII); it explains alerts in terms of simulated hemodynamics; and its alert rule generalized from development to held-out seeds. Continuous state makes the twin a persistent model of a patient rather than a fresh simulation each day.

Is the mechanistic layer circular? Partly. The Pulse scenario is selected by the classifier’s label and scaled by its severity, so the simulated physiology re-expresses the classifier’s decision through a body model rather than observing new evidence. What the engine contributes is a nonlinear, physiologically constrained mapping from that decision to hemodynamic stress: modest severities in benign scenarios produce no stress, and the risk score ignores them. The ablation shows this is where its value lies: a 7-fold reduction in false alerts at a cost of 30 points of sensitivity. The engine cannot recover a deterioration the classifier mislabels, and whether its explanations are interpretable to clinicians has not been tested with clinicians.

How personal is the twin? Less than the term usually implies. On real patients, every PerHeart body was an age proxy, ejection fraction and NT-proBNP were defaults, simulated cardiac output was 45–46% below measured in the two BCG patients, and the severity regressor relies on resting-HR and step trends for about 90% of its importance (EF: 0.6%). The current system is best described as a population-parameterized physiological model driven by an individual’s wearable trends; per-patient calibration of baseline hemodynamics is the main step toward a personal twin.

What it does not yet do. Three findings bound the claims. First, two of five deterioration types are invisible to the current alert because their signatures never reach the simulated hemodynamics: deconditioning by design (no acute event), and fluid overload because the scenario reduces venous compliance without adding circulating volume, which mobilizes pooled blood rather than overloading the circulation. Second, the models are sensitive to ordinary wearable noise and to clinical inputs outside their training distribution. Third, the twin’s own scores do not predict real outcomes beyond chance. Fourth, even the best configuration produces roughly one false alert every 11–12 days per stable patient, a burden that would be hard to sustain clinically. Our results therefore support feasibility and a specific mechanism (false-alarm filtering), not clinical effectiveness.

Lessons for building physiology-engine twins. Every major fault in Table VII was found by running the real system, not by offline evaluation: the NYHA skew only appeared in live inference, the CORS fault only in a browser, the resume drift only with the production modifier, and the alert mismatch only by auditing what the dashboard actually displayed. Pre-registering expectations and holding out seeds kept fixes honest, and reporting failure mechanisms rather than bare completion rates prevented a biased reading of the results.

IX. LIMITATIONS

Training data are synthetic, although every parameter is traced to a real aggregate statistic, and the pre-registered cohort is small (ten stories, six seeds). The fluid-overload scenario contains a modeling error rather than a mere limitation: it lowers venous compliance without adding circulating volume, which mobilizes pooled blood and moves the simulated hemodynamics in the improving direction as severity rises. Correcting it requires a volume-loading action, re-verification of engine stability, re-running the scenario validation, retraining the severity model and repeating the 21-day test; we have not yet done so. Several scoring elements were tuned post hoc (Table VI). Pulse’s native range (age 18–65, BMI 16–30) excludes most HF patients, so older patients are simulated with a proxy body. Real EF and NT-proBNP were unavailable in the replay cohort, and their defaults mask the fluid-overload mechanism. The engine’s SpO₂ output was unusable and its ECG is a template. Pulse is unstable when disease modifiers combine with sustained exertion, which under-represents high-severity exertion scenarios in all datasets. Medication effects are not modeled: Pulse includes a validated diuretic

TABLE XIX
SOFTWARE AND INFRASTRUCTURE ISSUES

Problem	Root cause (how found)	Solution	Evidence
Timeouts at 180 s	Some runs need ≈ 180 s under arm64 emulation	Documented margin; 2-worker cap	Re-attempts at 180.3–180.4 s
Failures under concurrency	SQLAlchemy pool (5+10) exhausted; CPU contention	Concurrency 9 $\rightarrow$ 4 $\rightarrow$ 2	0 failures at 2 workers
Corrupted Pulse image Python 3.9 container errors	Local containerd-snapshotter extraction bug PEP 604 unions evaluated at class definition by SQLAlchemy/Pydantic	Disabled snapshotter, re-pulled Optional[...] in model/schema files	Engine verified Imports succeed
Dashboard could not call API	Browser enforces CORS (invisible to curl/TestClient)	CORS middleware (local only)	Found in browser test
Model drift across machines	Unpinned scikit-learn/numpy versions	Exact dependency pins; model SHA-256 checked before each run	Zero version warnings

(furosemide), making diuretic modeling feasible, but has no beta-blocker or ACE-inhibitor/ARB substances. Risk-bucket cut-points and the deterioration-rate calibration constants are engineering choices, not clinically validated thresholds. The API has no authentication and is suitable only for research use.

X. CONCLUSION AND FUTURE WORK

We built and evaluated a wearable-driven physiological twin for HF that interprets home-measurable trends through a mechanistic engine and carries simulated state from day to day. On a pre-registered cohort, its physiology layer acted as a false-alarm filter, reducing false alerts about 7-fold relative to the classifier alone, while detecting 70% of deteriorating series and missing fluid overload and deconditioning; it ran end to end on real home data, but its scores did not predict real outcomes. Its failures are specific and explained, which makes them actionable. Future work will (1) add a volume-loading mechanism to the fluid-overload scenario and a diuretic action using Pulse’s furosemide model; (2) systematically map the engine’s stability boundary across severity, scenario and body profile; (3) make the models robust to wearable noise and out-of-distribution clinical inputs; (4) personalize the twin’s baseline heart rate and hemodynamics from measured data; and (5) evaluate day-by-day early warning on real longitudinal data with recorded HF events, the test that ultimately decides clinical value.

APPENDIX A

SOFTWARE AND INFRASTRUCTURE ISSUES

Table XIX lists issues that were general to the software stack rather than to the engine or the model; they are reported for reproducibility.

DATA AND CODE AVAILABILITY

Synthetic data, derived results and code are in the project repository. Raw MIMIC-IV, Zigong and PerHeart files are not redistributed, in line with their access agreements; PerHeart-derived outputs are shared under CC BY-NC-SA.

REFERENCES

- [1] S. L. James *et al.*, “Global, regional, and national incidence, prevalence, and years lived with disability for 354 diseases and injuries for 195 countries and territories, 1990–2017,” *Lancet*, vol. 392, pp. 1789–1858, 2018, doi: 10.1016/S0140-6736(18)32279-7.
- [2] G. Savarese *et al.*, “Global burden of heart failure: A comprehensive and updated review of epidemiology,” *Cardiovasc. Res.*, vol. 118, no. 17, pp. 3272–3287, 2022, doi: 10.1093/cvr/cvac013.
- [3] P. A. Heidenreich *et al.*, “2022 AHA/ACC/HFSA guideline for the management of heart failure,” *Circulation*, vol. 145, 2022, doi: 10.1161/CIR.0000000000001063.
- [4] S. C. Inglis *et al.*, “Structured telephone support or telemonitoring programs for patients with chronic heart failure,” *Cochrane Database Syst. Rev.*, 2008, doi: 10.1002/14651858.CD007228.
- [5] W. T. Abraham *et al.*, “Wireless pulmonary artery haemodynamic monitoring in chronic heart failure: A randomised controlled trial,” *Lancet*, vol. 377, pp. 658–666, 2011, doi: 10.1016/S0140-6736(11)60101-3.
- [6] J. Stehlik *et al.*, “Continuous wearable monitoring analytics predict heart failure hospitalization: The LINK-HF multicenter study,” *Circ. Heart Fail.*, vol. 13, Art. no. e006513, 2020, doi: 10.1161/CIRCHEARTFAILURE.119.006513.
- [7] N. Gautam *et al.*, “Artificial intelligence, wearables and remote monitoring for heart failure: Current and future applications,” *Diagnostics*, vol. 12, Art. no. 2964, 2022, doi: 10.3390/diagnostics12122964.
- [8] J. Corral-Acero *et al.*, “The ‘Digital Twin’ to enable the vision of precision cardiology,” *Eur. Heart J.*, vol. 41, pp. 4556–4564, 2020, doi: 10.1093/eurheartj/ehaa159.
- [9] G. Coorey *et al.*, “The health digital twin to tackle cardiovascular disease—A review of an emerging interdisciplinary field,” *npj Digit. Med.*, vol. 5, Art. no. 126, 2022, doi: 10.1038/s41746-022-00640-7.
- [10] S. A. Niederer *et al.*, “Scaling digital twins from the artisanal to the industrial,” *Nat. Comput. Sci.*, vol. 1, pp. 313–320, 2021, doi: 10.1038/s43588-021-00072-5.
- [11] F. Gu *et al.*, “Identification of digital twins to guide interpretable AI for diagnosis and prognosis in heart failure,” *npj Digit. Med.*, vol. 8, Art. no. 110, 2025, doi: 10.1038/s41746-025-01501-9.
- [12] J. Geddes *et al.*, “Digital twins for noninvasively measuring predictive markers of right heart failure,” *npj Digit. Med.*, vol. 8, Art. no. 545, 2025, doi: 10.1038/s41746-025-01920-8.
- [13] A. Balasubramanyam *et al.*, “A cardiovascular modeling framework for enabling personalized healthcare: A digital twin approach,” in *Proc. IEEE 13th Int. Conf. Healthcare Informatics (ICHI)*, 2025, pp. 478–489, doi: 10.1109/ICHI64645.2025.00062.
- [14] A. Bray *et al.*, “Pulse Physiology Engine: An open-source software platform for computational modeling of human medical simulation,” *SN Compr. Clin. Med.*, vol. 1, pp. 362–377, 2019, doi: 10.1007/s42399-019-00053-w.
- [15] S. J. Pocock *et al.*, “Predicting survival in heart failure: A risk score based on 39372 patients from 30 studies,” *Eur. Heart J.*, vol. 34, pp. 1404–1413, 2013, doi: 10.1093/eurheartj/ehs337.
- [16] C. van Walraven *et al.*, “Derivation and validation of an index to predict early death or unplanned readmission after discharge from hospital to the community,” *CMAJ*, vol. 182, no. 6, pp. 551–557, 2010, doi: 10.1503/cmaj.091117.

- [17] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.
- [18] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.
- [19] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.
- [20] A. E. W. Johnson *et al.*, "MIMIC-IV, a freely accessible electronic health record dataset," *Sci. Data*, vol. 10, Art. no. 1, 2023, doi: 10.1038/s41597-022-01899-x.
- [21] A. L. Goldberger *et al.*, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000, doi: 10.1161/01.CIR.101.23.e215.
- [22] D. Chicco and G. Jurman, "Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone," *BMC Med. Inform. Decis. Mak.*, vol. 20, Art. no. 16, 2020, doi: 10.1186/s12911-020-1023-5.
- [23] M. Kolakowski *et al.*, "Multimodal dataset of in-home physiological and inertial measurements from older heart failure patients," *Data*, vol. 11, Art. no. 106, 2026, doi: 10.3390/data11050106.
- [24] Z. Zhang *et al.*, "Electronic healthcare records and external outcome data for hospitalized patients with heart failure," *Sci. Data*, vol. 8, Art. no. 46, 2021, doi: 10.1038/s41597-021-00835-9.
- [25] J. Zhan *et al.*, "A multi-pathology ballistocardiogram dataset for cardiac function monitoring and arrhythmia assessment," *figshare*, 2025, doi: 10.6084/m9.figshare.28416896.
- [26] Royal College of Physicians, "National Early Warning Score (NEWS) 2: Standardising the assessment of acute-illness severity in the NHS," London, U.K., 2017.
- [27] J.-L. Vincent and D. De Backer, "Circulatory shock," *N. Engl. J. Med.*, vol. 369, no. 18, pp. 1726–1734, 2013, doi: 10.1056/NEJMr1208943.
- [28] A. Nohria, E. Lewis, and L. W. Stevenson, "Medical management of advanced heart failure," *JAMA*, vol. 287, no. 5, pp. 628–640, 2002, doi: 10.1001/jama.287.5.628.
- [29] D. A. Baran *et al.*, "SCAI clinical expert consensus statement on the classification of cardiogenic shock," *Catheter. Cardiovasc. Interv.*, vol. 94, pp. 29–37, 2019, doi: 10.1002/ccd.28329.